

# Where Prices Come From: The Interaction of Supply and Demand

ECON 101 - Macroeconomics

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- 3.1 The Demand Side of the Market
- 3.2 The Supply Side of the Market
- 3.3 Market Equilibrium: Putting Buyers and Sellers Together
- 3.4 The Effect of Demand and Supply Shifts on Equilibrium

## 3.1 Learning Objective

### Goal

Discuss the variables that influence demand.

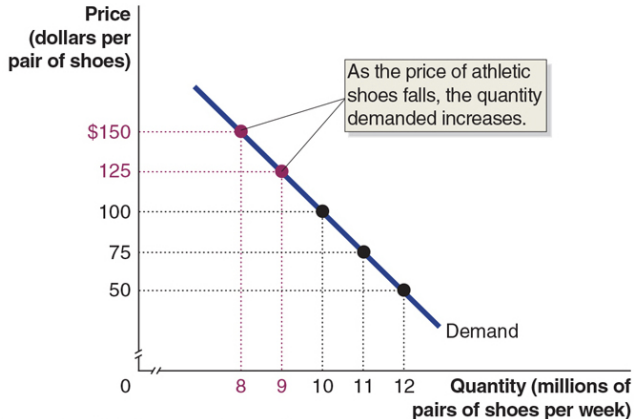
- We begin by investigating how **buyers** behave.
- **Market demand:** the demand by all the consumers of a given good or service.
- Key tools:
  - Demand schedule
  - Demand curve

## Figure 3.1 – A Demand Schedule and Demand Curve (1 of 3)

- **Demand schedule:** a table that shows the relationship between the **price** of a product and the **quantity of the product demanded**.
- **Demand curve:** a curve that shows the relationship between the price of a product and the quantity of the product demanded.

## Figure 3.1 – A Demand Schedule and Demand Curve

| Demand Schedule                         |                                                      |
|-----------------------------------------|------------------------------------------------------|
| Price<br>(dollars per<br>pair of shoes) | Quantity<br>(millions of pairs<br>of shoes per week) |
| \$150                                   | 8                                                    |
| 125                                     | 9                                                    |
| 100                                     | 10                                                   |
| 75                                      | 11                                                   |
| 50                                      | 12                                                   |



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## Figure 3.1 – Demand Curve and Ceteris Paribus (2 of 3)

- When drawing the demand curve, we assume **ceteris paribus**.
- **Ceteris paribus** (“all else equal”) condition:
  - When analyzing the relationship between two variables (such as price and quantity demanded), **other variables must be held constant**.

## Figure 3.1 – Quantity Demanded and the Law of Demand (3 of 3)

- **Quantity demanded:** the amount of a good or service that a consumer is willing and able to purchase at a given price.
- **Law of demand:**
  - Holding everything else constant, when the price of a product falls, the quantity demanded of the product will **increase**.
  - When the price of a product rises, the quantity demanded of the product will **decrease**.

# What Explains the Law of Demand?

- When the price of a good falls, two key effects occur:
  - **Substitution effect:** consumers substitute toward the good whose price has fallen.
  - **Income effect:** consumers have more purchasing power, which is like an increase in income.

# Substitution and Income Effects – Definitions

- **Substitution effect:**

- The change in the quantity demanded of a good that results from a change in price, making the good more or less expensive **relative to other goods**, holding constant the effect of the price change on consumer purchasing power.

- **Income effect:**

- The change in the quantity demanded of a good that results from the effect of a change in the good's price on consumers' **purchasing power**, holding all other factors constant.

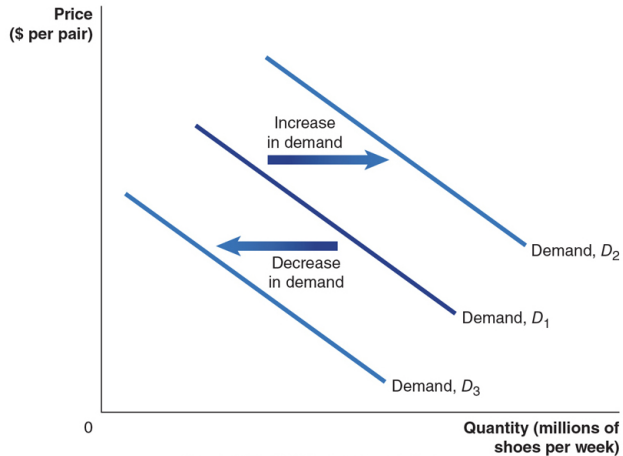
# Everyday Example: Movie Tickets

- Suppose movie ticket prices fall from \$15 to \$10.
- **Substitution effect:**
  - Movies are cheaper relative to other entertainment (e.g., bowling, dining out), so you substitute toward movies.
- **Income effect:**
  - You effectively have more real income; your budget goes further, so you can afford more movie nights per month.

## Figure 3.2 – Shifting the Demand Curve (1 of 2)

- A change in something **other than price** that affects demand causes the **entire demand curve to shift**.
- A shift to the right ( $D_1$  to  $D_2$ ) is an **increase in demand**.
- A shift to the left ( $D_1$  to  $D_3$ ) is a **decrease in demand**.

## Figure 3.2 – Shifting the Demand Curve



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## Figure 3.2 – Demand Shift vs. Quantity Demanded (2 of 2)

- As the demand curve shifts, the quantity demanded will change, **even if the price does not change**.
- The quantity demanded changes at **every possible price**.

# Variables That Shift Market Demand

- Income of consumers
- Prices of related goods
- Tastes
- Population and demographics
- Expectations

# Changes in Income of Consumers

- **Normal goods:** goods for which demand increases as income rises and decreases as income falls.
  - Examples: new clothes, restaurant meals, vacations.
- **Inferior goods:** goods for which demand increases as income falls and decreases as income rises.
  - Examples: second-hand clothes, instant noodles.

# Effects of Changes in Income

- An increase in income will:
  - Increase the demand for new clothes (normal good), *ceteris paribus*.
  - Decrease the demand for second-hand clothes (inferior good).

## Apply the Concept: Lobster as Inferior vs. Normal Good

- In some parts of the Maritimes, lobster is an **inferior good**: as incomes rise, people eat *less* lobster.
- In large cities, lobster is a **normal good**: as incomes rise, people eat *more* lobster.
- Key lesson:
  - Whether a good is normal or inferior depends on **consumer behaviour** and context, not its intrinsic characteristics.

# Changes in the Price of Related Goods

- **Substitutes:** goods and services that can be used for the same purpose.
  - Examples: Big Mac and Whopper; Ford F-150 and Dodge Ram; jeans and khakis.
- **Complements:** goods and services that are used together.
  - Examples: Big Mac and McDonald's fries; hot dogs and hot dog buns.

# Effects of Changes in the Price of Related Goods

- An increase in the price of a Big Mac:
  - **Increases** the demand for Whoppers (substitute).
  - **Decreases** the demand for McDonald's fries (complement).

# Changes in Tastes

- If consumer tastes change, they may buy more or less of a product.
- Example:
  - If consumers become more concerned about healthy eating, they might decrease their demand for fast food.

# Changes in Population and Demographics

- **Demographics:** characteristics of a population (age, race, gender, etc.).
- Increases in the **number of people** buying something will increase the amount demanded.
- Example:
  - An increase in the elderly population increases the demand for medical care.

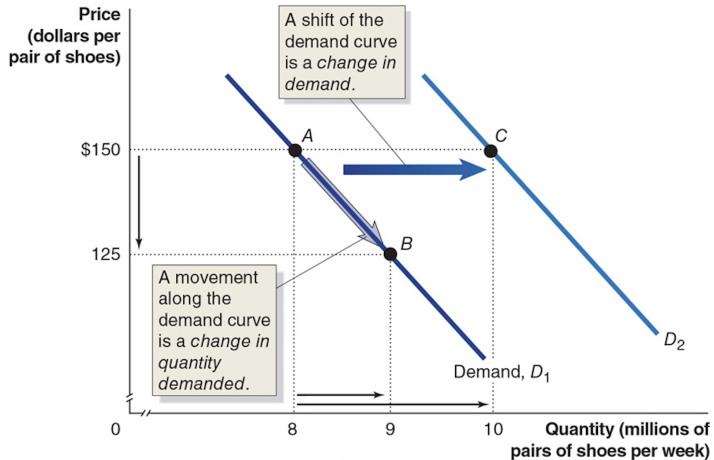
# Changes in Expectations About Future Prices

- Consumers decide not only **which** products to buy but also **when** to buy them.
- Future products are **substitutes** for current products.
- An expected **increase** in price tomorrow  $\Rightarrow$  increases demand today.
- An expected **decrease** in price tomorrow  $\Rightarrow$  decreases demand today.
- Example: If you find out that the price of gasoline will go up tomorrow, you increase your demand today.

## Figure 3.3 – Change in Demand vs. Change in Quantity Demanded

- A change in the **price of the product itself** causes a **movement along** the demand curve:
  - This is a **change in quantity demanded**.
- Any other change affecting demand causes the **entire demand curve to shift**:
  - This is a **change in demand**.

Figure 3.3



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## 3.2 Learning Objective

### Goal

Discuss the variables that influence supply.

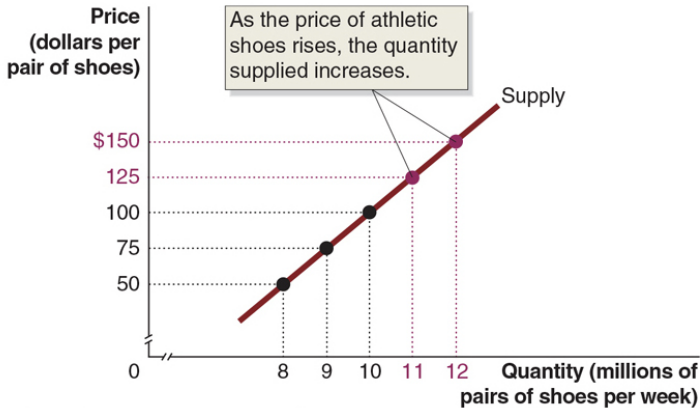
- There are similarities and important differences between the demand and supply sides.
- Now we examine **market supply**: decisions of firms about how much of a product to provide at various prices.

## Figure 3.4 – A Supply Schedule and Supply Curve (1 of 2)

- **Supply schedule:** a table that shows the relationship between the price of a product and the quantity of the product supplied.
- **Supply curve:** a curve that shows the relationship between the price of a product and the quantity of the product supplied.

## Figure 3.4 – Supply Schedule and Curve

| Supply Schedule                         |                                                      |
|-----------------------------------------|------------------------------------------------------|
| Price<br>(dollars per<br>pair of shoes) | Quantity<br>(millions of pairs<br>of shoes per week) |
| \$150                                   | 12                                                   |
| 125                                     | 11                                                   |
| 100                                     | 10                                                   |
| 75                                      | 9                                                    |
| 50                                      | 8                                                    |



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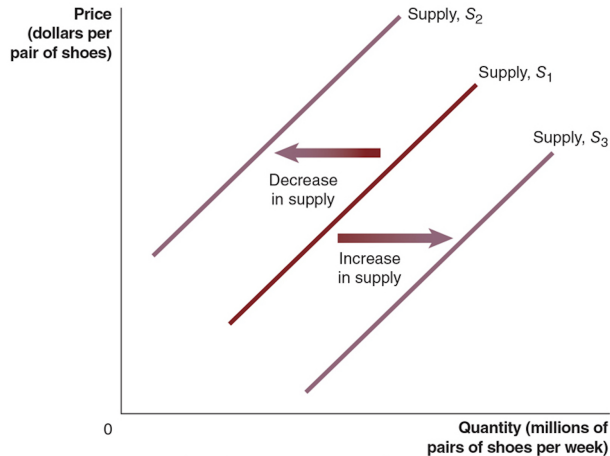
## Figure 3.4 – Quantity Supplied and Law of Supply (2 of 2)

- **Quantity supplied:** the amount of a good or service that a firm is willing and able to supply at a given price.
- **Law of supply:**
  - Holding everything else constant, increases in price cause **increases** in quantity supplied.
  - Decreases in price cause **decreases** in quantity supplied.

## Figure 3.5 – Shifting the Supply Curve (1 of 2)

- A change in something **other than price** that affects supply causes the entire supply curve to **shift**.
- A shift to the right ( $S_1$  to  $S_3$ ) is an **increase in supply**.
- A shift to the left ( $S_1$  to  $S_2$ ) is a **decrease in supply**.

## Figure 3.5 – Shifting the Supply Curve



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## Figure 3.5 – Supply Shift vs. Quantity Supplied (2 of 2)

- As the supply curve shifts, the quantity supplied changes **even if the price does not change**.
- The quantity supplied changes at every possible price.

# Variables That Shift Market Supply

- Prices of inputs
- Technological change
- Prices of substitutes in production
- Number of firms in the market
- Expected future prices

# Changes in Prices of Inputs

- **Inputs** are anything used in the production of a good or service.
  - For athletic shoes: rubber, plastic, and labour.
- An **increase** in the price of an input decreases the profitability of selling the good, causing a **decrease in supply**.
- A **decrease** in the price of an input increases profitability, causing an **increase in supply**.

- A firm may experience a positive or negative change in its ability to produce a given level of output with a given quantity of inputs.
- This is called **technological change**.
- Examples:
  - A new, more efficient way of producing shoes  $\Rightarrow$  **increases** supply.
  - Government restrictions on how much workers can work  $\Rightarrow$  **decreases** supply.

# Prices of Substitutes and Complements in Production

- Many firms can produce and sell alternative products: **substitutes in production**.
- Example:
  - An Ontario farmer can plant either corn or soybeans.
  - If the price of soybeans rises, the farmer will plant (supply) less corn.
- Sometimes two products are necessarily produced together: **complements in production**.
- Example:
  - Cattle provide both beef and leather.
  - An increase in the price of beef encourages more cattle farming, which increases the supply of leather.

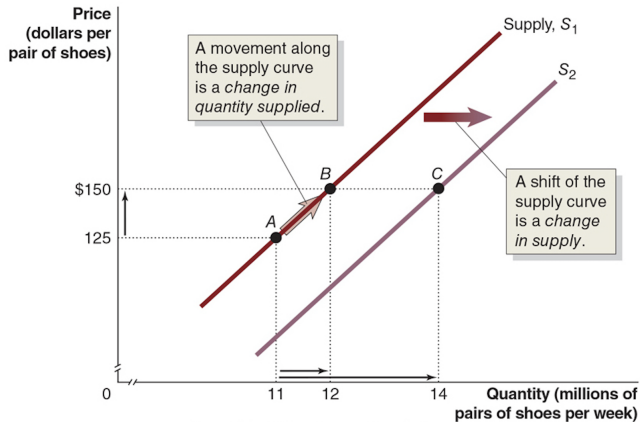
## Number of Firms and Expected Future Prices

- More firms in the market  $\Rightarrow$  more product available at a given price (greater supply).
- If a firm anticipates that the price of its product will be **higher** in the future, it might **decrease** its supply today to increase it in the future.

## Figure 3.6 – Change in Supply vs. Change in Quantity Supplied

- A change in the **price of the product itself** causes a **movement along** the supply curve:
  - This is a **change in quantity supplied**.
- Any other change affecting supply shifts the entire supply curve:
  - This is a **change in supply**.

Figure 3.6



## 3.3 Learning Objective

### Goal

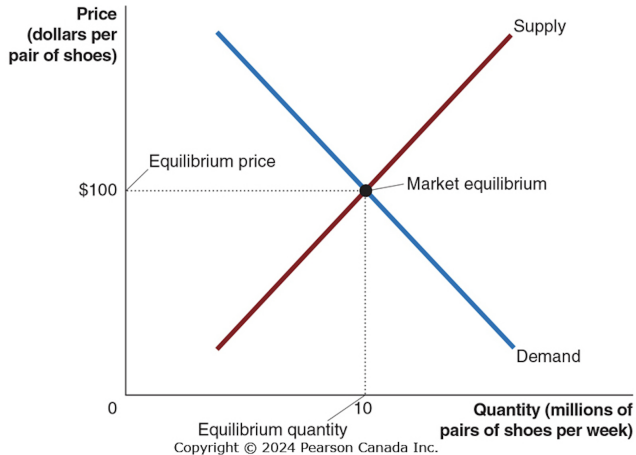
Use a graph to illustrate market equilibrium.

- **Market equilibrium:** quantity demanded equals quantity supplied.
- In markets with many buyers and sellers (perfectly competitive), we call this a **competitive market equilibrium**.

## Figure 3.7 – Market Equilibrium (Athletic Shoes)

- Consider the market for athletic shoes.
- At a price of \$100:
  - Consumers want to buy 10 million pairs of shoes per week.
  - Producers want to sell 10 million pairs of shoes per week.
- The **equilibrium price** is \$100, and the **equilibrium quantity** is 10 million pairs per week.

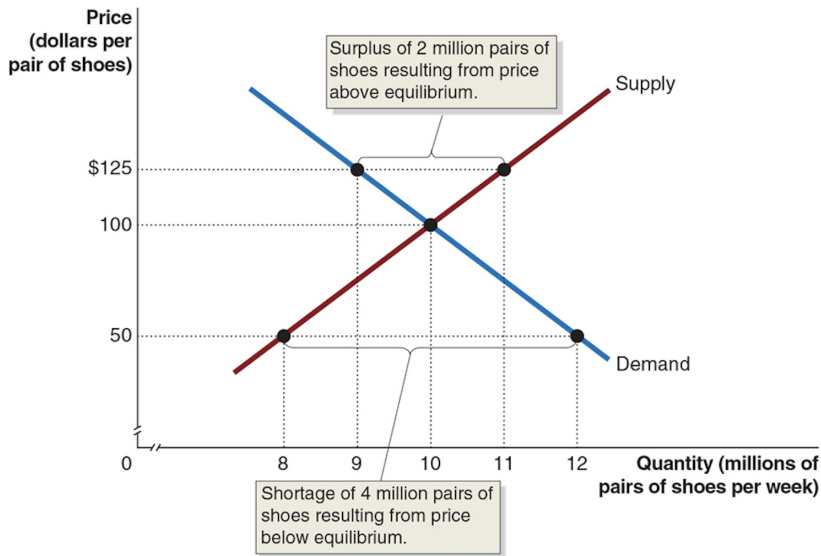
## Figure 3.7 – Market Equilibrium



## Figure 3.8 – Surpluses and Shortages (1 of 2)

- What if the price were \$125 instead?
- At \$125:
  - Consumers want to buy 9 million pairs of shoes.
  - Producers want to sell 11 million pairs of shoes.
- This gives a **surplus** of 2 million pairs: quantity supplied exceeds quantity demanded.
- Prediction: sellers will compete among themselves, driving the price **down** toward equilibrium.

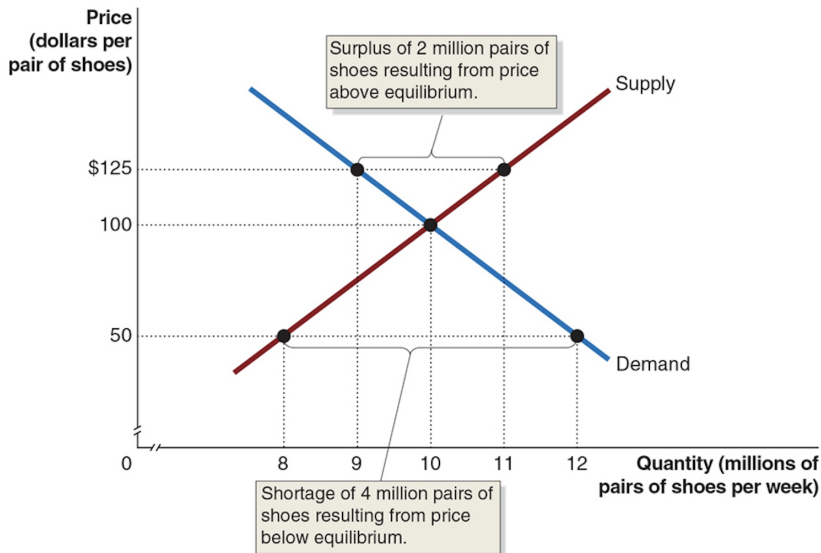
## Figure 3.8 – Surplus



## Figure 3.8 – Surpluses and Shortages (2 of 2)

- What if the price were \$50?
- At \$50:
  - Consumers want to buy 12 million pairs of shoes.
  - Producers want to sell 8 million pairs.
- This gives a **shortage** of 4 million pairs: quantity demanded exceeds quantity supplied.
- Prediction: sellers realize they can increase price and still sell as many pairs  $\Rightarrow$  price will **rise** toward equilibrium.

Figure 3.8 – Shortage



# Demand and Supply Both Count

- Price is determined by the **interaction** of buyers and sellers.
- In a competitive market, neither group alone can dictate the price.
- However, **changes in demand and/or supply** will affect the equilibrium price and quantity traded.

## 3.4 Learning Objective

### Goal

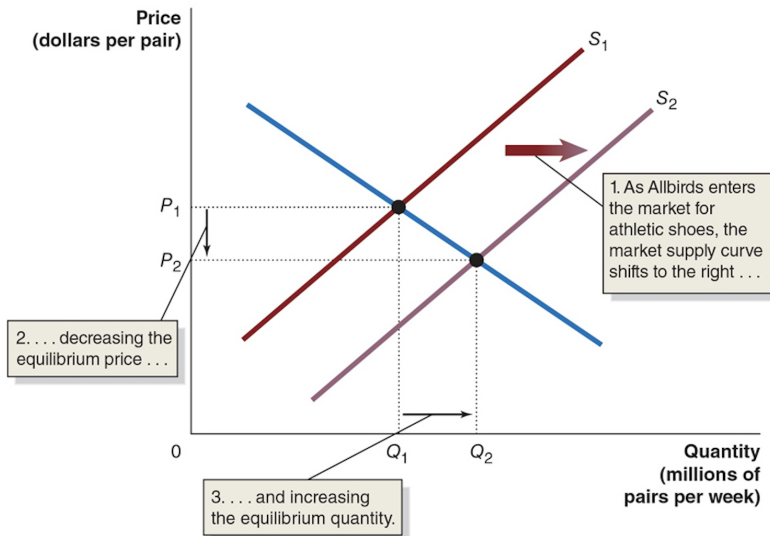
Use demand and supply graphs to predict changes in prices and quantities.

- We typically know the actual price and quantity, but do *not* observe the whole curves.
- The power of the model is in predicting **directional changes** in price and quantity when curves shift.

## Figure 3.9 – Effect of an Increase in Supply (1 of 2)

- Market for athletic shoes when a new shoemaker enters the market.
- At any given price, more shoes are supplied:
  - Supply shifts right from  $S_1$  to  $S_2$ .
- **Result:**
  - Equilibrium price **falls**.
  - Equilibrium quantity **rises**.

## Figure 3.9 – Increase in Supply



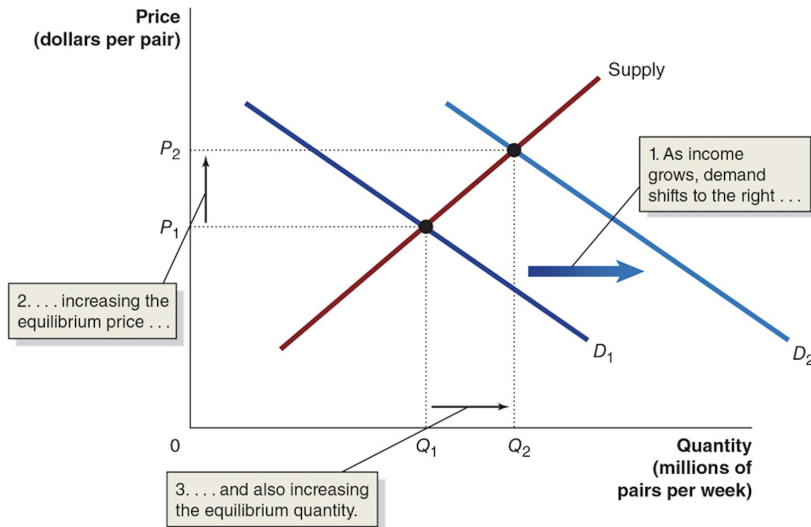
## Figure 3.9 – How Much Does Price Fall? (2 of 2)

- By how much will price fall?
- By how much will quantity rise?
- We **cannot** say without knowing more (the exact shapes and positions of demand and supply).
- For now, we only predict *directions*: price falls, quantity rises.

## Figure 3.10 – Effect of an Increase in Demand

- Suppose incomes increase.
- Athletic shoes are a **normal good**, so as income rises, demand shifts to the right.
- **Result:**
  - Equilibrium price **rises**.
  - Equilibrium quantity **rises**.

## Figure 3.10 – Increase in Demand



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## Table 3.3 – How Shifts in Demand and Supply Affect Equilibrium $P$ and $Q$

(1 of 2)

|                                             | Equilibrium Price $P$ | Equilibrium Quantity $Q$ |
|---------------------------------------------|-----------------------|--------------------------|
| <b>Increase in demand, supply unchanged</b> | Rises                 | Rises                    |
| <b>Decrease in demand, supply unchanged</b> | Falls                 | Falls                    |
| <b>Increase in supply, demand unchanged</b> | Falls                 | Rises                    |
| <b>Decrease in supply, demand unchanged</b> | Rises                 | Falls                    |

## Table 3.3 – How Shifts in Demand and Supply Affect $P$ and $Q$ (2 of 2)

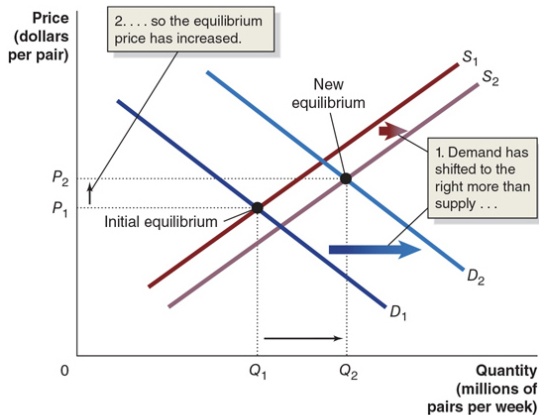
|                                               | Equilibrium Price $P$ | Equilibrium Quantity $Q$ |
|-----------------------------------------------|-----------------------|--------------------------|
| <b>Increase in supply, demand unchanged</b>   | Falls                 | Rises                    |
| <b>Increase in demand, decrease in supply</b> | Rises                 | ?                        |
| <b>Decrease in demand, increase in supply</b> | Falls                 | ?                        |
| <b>Increase in demand, increase in supply</b> | ?                     | Rises                    |
| <b>Decrease in demand, decrease in supply</b> | ?                     | Falls                    |

- The highlighted row (in blue text) is the example we studied: **increase in supply**.
- “?” means the effect is **ambiguous** without more information on the relative size of shifts.

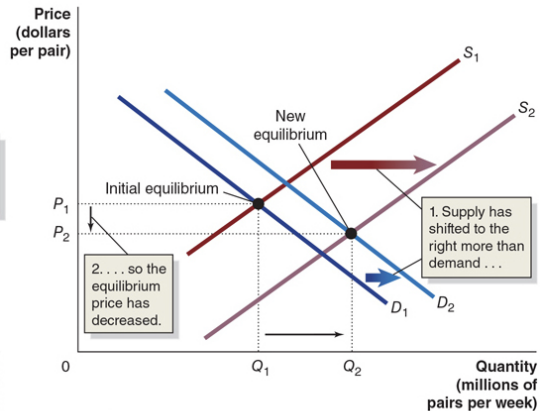
## Figure 3.11 – Shifts in Demand and Supply Over Time (1 of 3)

- Over time, it is likely that **both** demand and supply will change.
- Example:
  - New firms enter  $\Rightarrow$  supply shifts right.
  - Incomes rise  $\Rightarrow$  demand shifts right.

## Figure 3.11 – Both Demand and Supply Shift



(a) Demand shifting more than supply



(b) Supply shifting more than demand

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## Figure 3.11 – Case 1: Demand Shifts More than Supply (2 of 3)

- When demand shifts more than supply:
  - Equilibrium quantity **rises**.
  - Equilibrium price also **rises**.
- This panel shows a larger rightward shift of demand than of supply.

## Figure 3.11 – Case 2: Supply Shifts More than Demand (3 of 3)

- When supply shifts more than demand:
  - Equilibrium quantity **rises**.
  - Equilibrium price **falls**.
- Without knowing the relative size of the changes, the effect on price is ambiguous.
- It is possible, but unlikely, that equilibrium price remains unchanged.

# Shifts of Curves vs. Movements Along Curves

- Suppose an **increase in supply** occurs.
- We know:
  - Equilibrium quantity will increase.
  - Equilibrium price will decrease.
- It is tempting to say “the decrease in price causes an increase in demand”, but this is incorrect.
- The decrease in price causes a **movement along the demand curve**, not a shift of the curve.
- The demand curve already describes how much consumers want to buy at **any given price**.

# Summary

- Demand:
  - Follows the law of demand; explained by substitution and income effects.
  - Shifts when income, prices of related goods, tastes, population, or expectations change.
- Supply:
  - Follows the law of supply.
  - Shifts when input prices, technology, related goods in production, number of firms, or expected future prices change.
- Market equilibrium:
  - Occurs where quantity demanded equals quantity supplied.
  - Surpluses cause price to fall; shortages cause price to rise.
- The demand and supply model predicts **directional** changes in equilibrium price and quantity when curves shift.

Thank you!